

Lyapunov Exponents as Predictability Diagnostics for Differentiable Simulators and Learned World Models

Benjamin Strittmatter

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Abstract

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Evidence status. Figures 3, 5, and 8 use the corrected acrobot or its saved model study. Figures 1, 2, 4, 6, and 7 retain the earlier velocity-state acrobot experiment and are labelled historical; their acrobot results do not validate the corrected simulator.

1 Introduction

[To be written last.]

2 Background

2.1 Lyapunov exponents and λ_1

Chaos is a concept borrowed from information theory, which quantifies how inherently difficult it is to make predictions about the future of dynamical systems. Mathematically speaking, a hallmark of chaos is the exponential divergence of nearby trajectories:

$$x(t) - \tilde{x}(t) \sim e^{\lambda t}, \quad (1)$$

where $\tilde{x}(t) = x(t) + \delta(t)$ is related to the original trajectory by a small perturbation $\delta(t)$. The exponential rate of growth is called the "Lyapunov exponent". Goal of this work is to demonstrate that the Lyapunov exponent plays an important role in characterising the predictability horizons in robotic learning.

To obtain a language that is more closely related to the robotics framework we want to study, it is advantageous to work in the tangent space picture. In practice, the trajectory $x(t)$ is determined by an ODE $\dot{x} = f(x)$, for some function $f(x(t))$. We can linearise the ODE for a perturbation $x + \delta$, and arrive at the expression

$$\dot{\delta} = Df(x) \delta, \quad (2)$$

with $Df(x)$ the Jacobian matrix. Mirroring the mechanics of a differentiable simulator, we discretise the flow with a fixed step dt . This gives a one-step map $x_{t+1} = F(x_t)$, whose linearisation propagates the perturbation by the one-step Jacobian J_t :

$$\delta_{t+1} = J_t \delta_t, \quad J_t = \frac{\partial x_{t+1}}{\partial x_t}. \quad (3)$$

Solving the discrete recursion gives $\delta_t = M_t \delta_0$, propagated by the *rollout Jacobian*

$$M_t = \frac{\partial x_t}{\partial x_0} = \prod_{s < t} J_s, \quad (4)$$

the fundamental matrix of the linearised flow. The *largest Lyapunov exponent* is the asymptotic growth rate of a generic infinitesimal perturbation,

$$\lambda_1 = \lim_{t \rightarrow \infty} \frac{1}{t} \ln \frac{\|M_t \delta_0\|}{\|\delta_0\|}, \quad (5)$$

a limit that exists for almost every initial condition by Oseledets' multiplicative ergodic theorem [1]. The order of limits matters— δ_0 infinitesimal first, then $t \rightarrow \infty$ —which is why the exponent is a property of the linearised flow rather than of two finite trajectories, whose separation saturates once

it is no longer small. A positive λ_1 is the operational signature of chaos: an initial uncertainty ε grows to order unity within the *predictability time* $\tau \sim \lambda_1^{-1} \ln(1/\varepsilon)$; the Lyapunov time $1/\lambda_1$ sets the amplification timescale, while the horizon also depends on initial uncertainty and the tolerated error. This work investigates how the *same* sensitivity growth informs two computational objects of modern robot learning—differentiable simulators (Part A) and learned world models (Part B).

The perturbation directions that grow fastest are the leading singular vectors of M_t ; its singular values $\sigma_i(M_t) \sim e^{\lambda_i t}$ define the full *Lyapunov spectrum* $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$. (Singular rather than eigenvalues: M_t is non-normal, and a pure shear has unit eigenvalues yet still stretches space.) The sum of the spectrum measures the mean exponential growth of phase-space volume [2],

$$\sum_{i=1}^n \lambda_i = \langle \nabla \cdot f \rangle = \frac{1}{dt} \langle \ln |\det J| \rangle, \quad (6)$$

the time-averaged divergence of the flow. The systems we study are Hamiltonian, so Liouville’s theorem gives $\nabla \cdot f = 0$ and the spectrum sums to zero: phase-space volume is conserved. This volume-preservation constraint is a structural fingerprint that a faithful model must reproduce—a theme we return to in Part B (Section 5)—and serves throughout as a consistency check on the estimator.

Reading the *full spectrum* off an explicitly formed M_t is numerically hopeless: the dynamic range $\sigma_1/\sigma_n \sim e^{(\lambda_1 - \lambda_n)t}$ over- or under-flows and every column collapses onto the dominant direction, burying the sub-dominant exponents within a few thousand steps. We instead use the standard Benettin/QR estimator [3, 4]: evolve an orthonormal frame Q under the one-step Jacobians, re-orthonormalise at each step through a factorisation $J_t Q = Q' R$, and accumulate the logarithms of the diagonal entries of R ; the running averages converge to the exponents λ_i (reported per unit time, after dividing by dt), with convergence guaranteed by Oseledets. A single such estimator is used throughout, calibrated against known answers: it returns zero to machine precision on a harmonic oscillator and recovers $\lambda_1 \approx 0.906$ for the Lorenz system, as detailed in Section 3.2.

2.2 Jacobians and differentiable simulation

A *differentiable simulator* is a physics engine whose one-step map $x_{t+1} = F(x_t)$ is implemented so that derivatives propagate through it: applying reverse-mode automatic differentiation to a rollout returns the gradient of a trajectory objective with respect to the initial state, the physical parameters, or a sequence of control inputs. We use NVIDIA Warp [5] for every simulator in this work; the implementation details are deferred to Section 3.1.

The object such a simulator differentiates is exactly the rollout Jacobian of Section 2.1. For a scalar loss $L(x_T)$ evaluated at the end of a T -step rollout, the chain rule gives

$$\frac{\partial L}{\partial x_0} = \frac{\partial L}{\partial x_T} M_T, \quad M_T = \prod_{t < T} J_t, \quad (7)$$

with M_T the rollout Jacobian of Eq. (4). Reverse-mode differentiation evaluates this product right-to-left, multiplying in one J_t per step as it walks back along the stored trajectory. Every gradient a differentiable simulator produces is therefore a product of one-step Jacobians—the same product that governs the growth of perturbations.

This is not a peculiarity of one loss. The three canonical uses of a differentiable simulator all backpropagate through the rollout and so all carry the factor M_T : system identification differentiates

the trajectory with respect to physical parameters, trajectory optimisation differentiates it with respect to a control sequence, and first-order (analytic) policy gradients differentiate it with respect to policy parameters [6, 7]. In each case the gradient inherits whatever M_T does.

What M_T does is set asymptotically by λ_1 . From the definition of the largest Lyapunov exponent, Eq. (5), the rollout Jacobian has leading growth $\|M_T\| \sim e^{\lambda_1 T}$ (up to a subexponential prefactor, all that survives when $\lambda_1 = 0$; Section 4.1). On a chaotic system ($\lambda_1 > 0$), this amplification can make gradients sensitive to small changes in initial conditions—the *curse of chaos* for differentiable simulators [8, 7, 6]. Part A measures the gradient-gain growth rate and explores the resulting optimisation difficulty. The useful rollout length also depends on the objective, perturbation scale, and optimisation budget; $1/\lambda_1$ is an amplification timescale rather than a universal cutoff.

Equation (7) also gives a second, independent handle on λ_1 . By Eq. (5), the growth rate of the gradient gain $\|M_T\|$ with T is itself a finite-horizon estimate of λ_1 —distinct from the Benettin/QR estimator of Section 2.1, computed from the same orbit, and expected to agree with it as $T \rightarrow \infty$ by Oseledets. Section 3.3 makes this comparison the basis of Part A.

2.3 The systems

We study three systems that span the range of dynamical behaviour we care about: one integrable and two chaotic of contrasting character. Each is a conservative Hamiltonian system: writing the state as a configuration q with conjugate momentum p , the dynamics follow from a Hamiltonian

$$H(q, p) = \frac{1}{2} p^\top M(q)^{-1} p + V(q), \quad (8)$$

the sum of kinetic and potential energy, with $M(q)$ the configuration-dependent mass (inertia) matrix [9]. Equivalently, in second-order form the passive dynamics obey the manipulator equation $M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = 0$ [10]; control enters as a forcing term on the right, but our Lyapunov diagnostics concern the underlying conservative flow, for which Liouville’s theorem applies and the spectrum sums to zero as in Section 2.1.

The three systems form a deliberate contrast in λ_1 :

- **Pendulum** — a single degree of freedom with separable Hamiltonian $H = \frac{1}{2}p^2 + V(\theta)$. A one-degree-of-freedom autonomous Hamiltonian is integrable: energy conservation confines every orbit to a level set of H , leaving no room for exponential divergence, so $\lambda_1 = 0$ [9]. This is our ordered baseline.
- **Acrobot** — a two-link pendulum actuated only at the elbow, the textbook chaotic under-actuated system [11]. Its inertia matrix $M(q)$ depends on the configuration, making the Hamiltonian *non-separable*, and its passive dynamics admit chaotic regimes, $\lambda_1 > 0$.
- **Hénon–Heiles** — a unit-mass particle in the smooth two-dimensional potential $V(x, y) = \frac{1}{2}(x^2 + y^2) + x^2y - \frac{1}{3}y^3$, a classical model from galactic dynamics [12]. Its Hamiltonian is *separable* (constant mass matrix $M = I$), yet for energies approaching the escape value $E = \frac{1}{6}$ the motion is strongly chaotic, $\lambda_1 > 0$. It is a second, mechanistically independent chaotic system—of different origin and structure from the acrobot—included to show the gradient diagnostics track chaos itself, not the mechanical setting. Its phase space is *mixed* (regular tori coexist with a chaotic sea), so we launch orbits in the chaotic sea; λ_1 is $O(0.1)$, small but genuinely positive.

The separable/non-separable distinction is the structural divide. When M is constant the kinetic term in Eq. (8) decouples from q , and the position and momentum updates can be taken independently; when $M(q)$ varies—as for the acrobot—the two are coupled. That coupling is the reason it is the demanding case for the structure-preserving integrators and world models of Part B. Chaos does not require it, however: Hénon–Heiles is separable ($M = I$) yet chaotic, its sensitivity arising from the anharmonic potential rather than a configuration-dependent inertia.

Because the consistency checks of Section 2.1 rely on volume preservation, the integrator must preserve the Hamiltonian structure. We use semi-implicit (symplectic) Euler for the separable pendulum and Hénon–Heiles, and canonical implicit midpoint for the non-separable acrobot [13, 14]. These symplectic maps preserve phase-space volume: their one-step Jacobians satisfy $\det J = 1$, implying $\sum_i \lambda_i = 0$. For implicit midpoint, this statement holds for the exactly solved step; our fixed-iteration implementation is checked against its analytic Jacobian. The earlier velocity-first acrobot scheme was not symplectic and is retained only in historical results. The step kernels are detailed in Section 3.1. The resulting ladder— $\lambda_1 = 0$ for the pendulum up to $\lambda_1 > 0$ for chaotic acrobot and Hénon–Heiles regimes—is the controlled axis along which Part A sweeps the gradient diagnostics and Part B contrasts a learned model’s recovered exponent against the truth.

3 Methods — what we simulate and how

3.1 Differentiable Warp simulators

We implement each system of Section 2.3 directly as a Warp [5] compute kernel. For the pendulum and Hénon–Heiles, it advances the state by one *velocity-first semi-implicit (symplectic) Euler* step: the acceleration is evaluated at the current state, the velocity is updated first, and the configuration is then advanced with the *new* velocity. Here ω denotes the velocity components: angular velocity for the pendulum and Cartesian velocities for Hénon–Heiles:

$$\omega_{t+1} = \omega_t + dt a(q_t, \omega_t), \quad q_{t+1} = q_t + dt \omega_{t+1}. \quad (9)$$

For these separable systems the map is symplectic. The acrobot instead uses canonical implicit midpoint, as described below. The underlying accelerations are:

- **Pendulum** (state (θ, ω)): $a = -(g/\ell) \sin \theta$.
- **Acrobot**: in angular-velocity coordinates $(\theta_1, \theta_2, \omega_1, \omega_2)$, the accelerations solve the manipulator equation $M(\theta) \dot{\omega} = F(\theta, \omega)$ with the point-mass double-pendulum inertia matrix and passive (Coriolis + gravity) forcing

$$\begin{aligned} M(\theta) &= \begin{bmatrix} (m_1 + m_2) l_1^2 & m_2 l_1 l_2 \cos(\theta_1 - \theta_2) \\ m_2 l_1 l_2 \cos(\theta_1 - \theta_2) & m_2 l_2^2 \end{bmatrix}, \\ F &= \begin{bmatrix} -m_2 l_1 l_2 \omega_2^2 \sin(\theta_1 - \theta_2) - (m_1 + m_2) g l_1 \sin \theta_1 \\ m_2 l_1 l_2 \omega_1^2 \sin(\theta_1 - \theta_2) - m_2 g l_2 \sin \theta_2 \end{bmatrix}. \end{aligned} \quad (10)$$

The kernel uses canonical coordinates (q, p) , with $q = (\theta_1, \theta_2)$ and $p = M(q)\omega$, Hamiltonian (8), and $V = -(m_1 + m_2)gl_1 \cos \theta_1 - m_2 gl_2 \cos \theta_2$. It advances Hamilton’s equations, $\dot{q} = \partial H / \partial p$ and $\dot{p} = -\partial H / \partial q$, using implicit midpoint: derivatives are evaluated at the midpoint between the current and next states. The implicit step is approximated with six fixed-point iterations. Control adds the applied torque to $\dot{p}_2$. Because M depends on θ , the Hamiltonian is non-separable, which is the structural feature exploited in Part B (Section 5).

- **Hénon–Heiles** (state (x, y, p_x, p_y)): a unit-mass particle in the smooth potential $V(x, y) = \frac{1}{2}(x^2 + y^2) + x^2y - \frac{1}{3}y^3$, so the accelerations are just the force $-\nabla V$, namely $\ddot{x} = -(x + 2xy)$ and $\ddot{y} = -(y + x^2 - y^2)$. The mass matrix is the identity, so unlike the acrobot the Hamiltonian is *separable* and the semi-implicit step is exactly symplectic (Section 2.3); the chaos here comes from the anharmonic potential, not a configuration-dependent inertia.

Reverse-mode differentiation requires the full trajectory to be available on the backward pass, so each rollout is stored as a $(T+1) \times \text{dim}$ array, writing one new row per step and never overwriting a cell; the adjoint then walks back along the stored states, multiplying in one one-step Jacobian J_t per step exactly as in Eq. (7). All simulators run on CPU, single-threaded. The integration step and physical parameters are collected in Table 1.

System	dt	Parameters
Pendulum	5×10^{-3}	$g = 9.81, \ell = 1$
Acrobot	5×10^{-4}	$m_1 = m_2 = 1, l_1 = l_2 = 1, g = 9.81$
Hénon–Heiles	10^{-2}	unit mass, standard coupling ($\lambda = 1$)

Table 1: Integration step and physical parameters of the three Warp simulators.

3.2 The Lyapunov estimator and its calibration

We operationalize the Benettin/QR estimator of Section 2.1 as follows. From a seeded random orthonormal frame $Q \in \mathbb{R}^{n \times k}$ (the k leading exponents; $k = n$ for the full spectrum), we advance the tangent frame one step at a time, re-orthonormalising through a thin QR factorisation $J_t Q = Q' R$ at *every* step, accumulating $\ln|R_{ii}|$, and dividing the running sums by the elapsed time $N dt$ to obtain the exponents in physical (per-unit-time) units, sorted in descending order (the telescoping that makes $\frac{1}{N dt} \sum_t \ln R_{ii}^{(t)} \rightarrow \lambda_i$ exact is spelled out in Appendix A). The initial transient is discarded by slicing the trajectory before the average: for the dissipative Lorenz calibration this is the relaxation onto the strange attractor (the first 2000 steps). The QR frame is initialised after slicing, so this does not provide a QR-alignment burn-in (Section 3.3). A single such estimator is used for every result in this paper.

We calibrate it against two systems with known answers.

- **Harmonic oscillator** (an exact rotation map, all exponents zero): the estimator returns $|\lambda_i| \lesssim 2 \times 10^{-15}$ over 2×10^4 steps—zero to machine precision, as expected since each step contributes $\ln|\pm 1| = 0$ exactly.
- **Lorenz** ($\sigma = 10, \rho = 28, \beta = 8/3$; RK4 at $dt = 5 \times 10^{-3}$, first 2000 steps dropped): the finite-time largest exponent is itself a fluctuating quantity. Single-trajectory estimates scatter by ~ 0.02 around the standard literature value $\lambda_1 = 0.9056$ over a few hundred Lyapunov times, and the running average on a long trajectory converges to it ($\lambda_1 = 0.905$ at 1.6×10^3 Lyapunov times). This scatter is intrinsic finite-time variance, not a Jacobian or estimator bias: finite-difference and analytic (variational-RK4) Jacobians agree on λ_1 to 3×10^{-8} along a common orbit. As a sharp structural check valid at *every* horizon, the recovered spectrum has a near-zero middle exponent and sums to $\sum_i \lambda_i = -13.667$, matching the constant flow divergence $\text{tr } Df = -(\sigma + 1 + \beta)$ to machine precision (Eq. (6)).

Finally, the per-step Jacobian feeding the estimator is computed analytically for the pendulum and Hénon–Heiles and each is cross-checked against Warp’s reverse-mode autodiff. The analytic

pendulum Jacobian agrees with autodiff to 1.7×10^{-9} and the Hénon–Heiles Jacobian to 1×10^{-4} . For the acrobot, we use the analytic Jacobian of the implicit midpoint step. Across the tested states and applied torques, its maximum absolute entrywise difference from Warp autodiff is below 10^{-6} at the production time step $dt = 5 \times 10^{-4}$. This verifies, within that tolerance, that the tangent map used by the Lyapunov estimator matches the implemented simulator step.

3.3 Two estimators of λ_1

Part A rests on comparing two finite-horizon estimates of λ_1 computed from the *same* orbit. The first is the calibrated Benettin/QR estimator of Section 3.2. The second reads λ_1 off the growth of the gradient gain introduced in Section 2.2: since $\|M_T\| \sim e^{\lambda_1 T}$, the least-squares slope of $\ln \|M_T\|_2$ against T is itself an estimate of λ_1 . Here $\|M_T\|_2$ is the spectral norm (top singular value) of the rollout Jacobian, and the slope is fit over a range of horizons reaching several Lyapunov times (T from $\sim 10^3$ up to 1.6×10^4 integration steps).

These two numbers are distinct finite- T functionals of the same Jacobian sequence, *not* algebraically identical: the QR estimator averages the local logarithmic stretching of an orthonormal frame, whereas the slope tracks the global spectral norm of the accumulated product. Both converge to the true λ_1 as $T \rightarrow \infty$ by Oseledets’ theorem, but only once $T \gg 1/(\lambda_1 - \lambda_2)$ —the time a generic direction needs to align with the leading one—and the slope is the higher-variance of the two. Part A therefore reports their agreement as a *consistency relation*—evidence that the gradient-gain growth rate and the dynamical λ_1 are one and the same number—rather than as an exact identity.

4 Part A — λ_1 bounds differentiable-simulator gradients

Part A establishes the first pillar of the thesis: the largest Lyapunov exponent λ_1 governs the gradients a differentiable simulator produces. Section 2.2 showed that every such gradient carries the rollout-Jacobian factor M_T with $\|M_T\| \sim e^{\lambda_1 T}$, and Section 3.3 set up a second, gradient-side estimate of λ_1 read off that growth rate.

Three systems play distinct roles. The chaotic *acrobot* and *Hénon–Heiles* are the workhorses and carry the λ_1 claims; the integrable *pendulum* ($\lambda_1 = 0$) is the null contrast, whose gradient grows only linearly, never exponentially. Hénon–Heiles, with a modest $\lambda_1 \approx 0.13$ close in magnitude to the pendulum’s finite-time floor, is the sharpest test that the diagnostics track chaos itself rather than merely a large exponent.

Against this cast we (i) illustrate the growth law and finite-window estimates (Section 4.1); (ii) test the relation across an acrobot energy sweep, the pendulum, and Hénon–Heiles (Section 4.2); (iii) examine analytic-gradient signal-to-noise across perturbed starts (Section 4.3); and (iv) compare precise reaching and forgiving swing-up under a fixed optimisation budget (Section 4.4). The growth-law and SNR plots retain the historical acrobot protocol; the regime sweep and optimisation comparison use the corrected simulator.

4.1 The gradient-gain growth law (pre-correction)

The rollout-Jacobian factor $\|M_T\|_2$ that every simulator gradient carries (Section 2.2) has a leading exponential form $\|M_T\|_2 \sim e^{\lambda_1 T}$; how it grows once that factor degenerates is precisely the integrable–chaotic distinction (Fig. 1). On the chaotic acrobot the gain is *exponential*: a log-linear fit past the initial transient gives 1.29 s^{-1} , matching the Benettin exponent $\lambda_1 = 1.28 \text{ s}^{-1}$ measured independently

Rollout-Jacobian gain $\|M_T\|_2$: linear (integrable) vs exponential (chaotic)

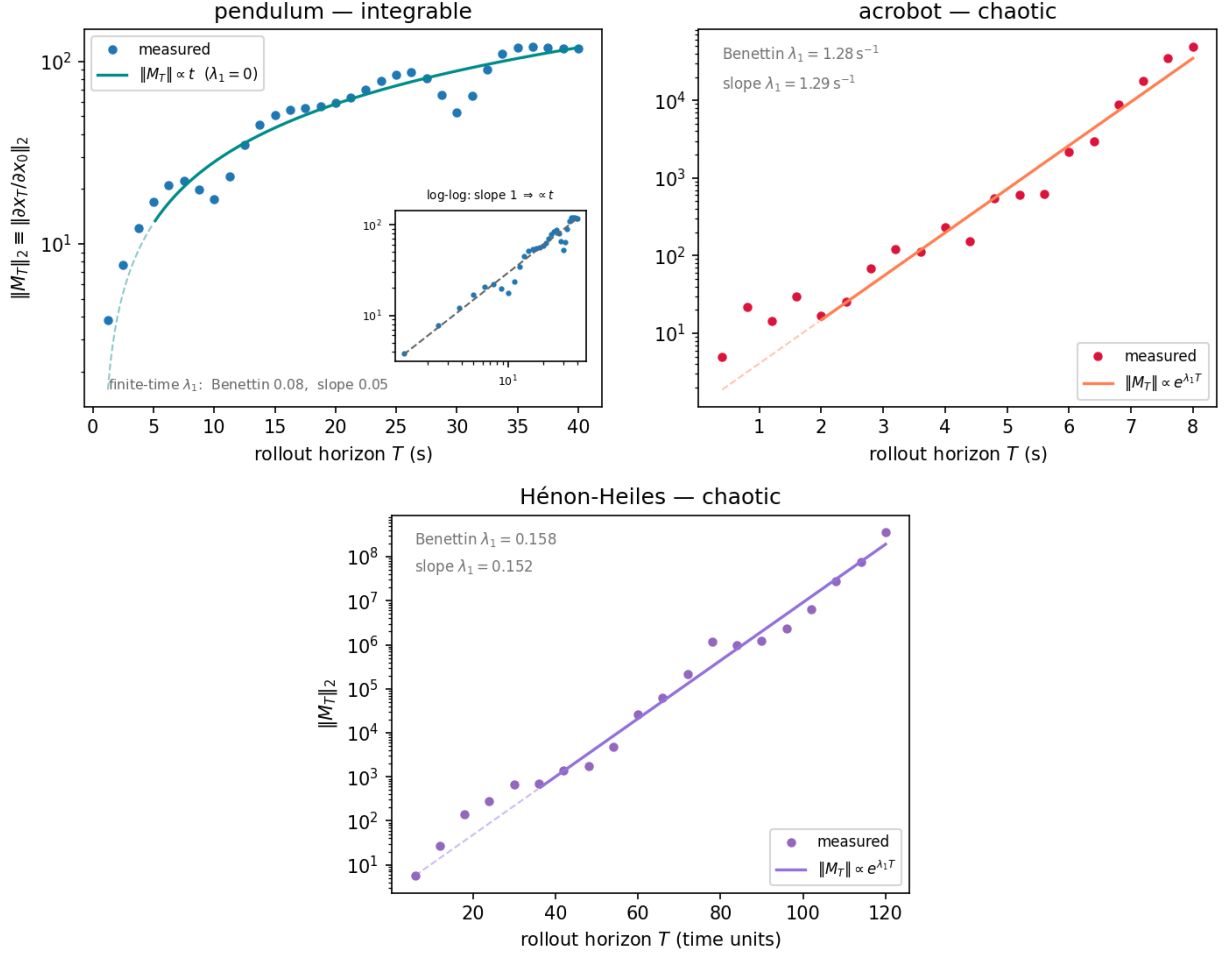


Figure 1: **Pre-correction.** Gradient gain $\|\partial x_T / \partial x_0\|_2$ versus rollout horizon T (semilog), each panel over its own time window. Top-left, integrable pendulum: *linear* growth $\|M_T\|_2 \propto T$ with $\lambda_1 = 0$ (inset: log-log slope 1); the quoted finite-time λ_1 are window artifacts that decay to 0. Top-right, chaotic acrobot: *exponential* growth, log-slope 1.29 s^{-1} matching the same-orbit Benettin $\lambda_1 = 1.28 \text{ s}^{-1}$. Bottom, chaotic Hénon-Heiles: likewise *exponential* at a smaller rate, log-slope 0.152 against Benettin 0.158 (dimensionless units). Fits are taken past the initial transient (solid; dashed extrapolation).

on the same orbit. Hénon-Heiles behaves identically at its own, smaller rate—the fit gives 0.152 against a same-orbit Benettin $\lambda_1 = 0.158$, a straight exponential over some eight decades of $\|M_T\|_2$. On the integrable pendulum the gain is *linear*, $\|M_T\|_2 \propto T$ with $\lambda_1 = 0$ exactly—the log-log inset shows the unit power law. The gradient gain thus blows up at the Lyapunov rate on chaotic dynamics, and grows only *polynomially*, never exponentially, on integrable ones.

The linear law is a property of integrable motion, not an accident of this orbit. Every orbit of the pendulum is periodic, and because the period depends on amplitude, neighbouring orbits run

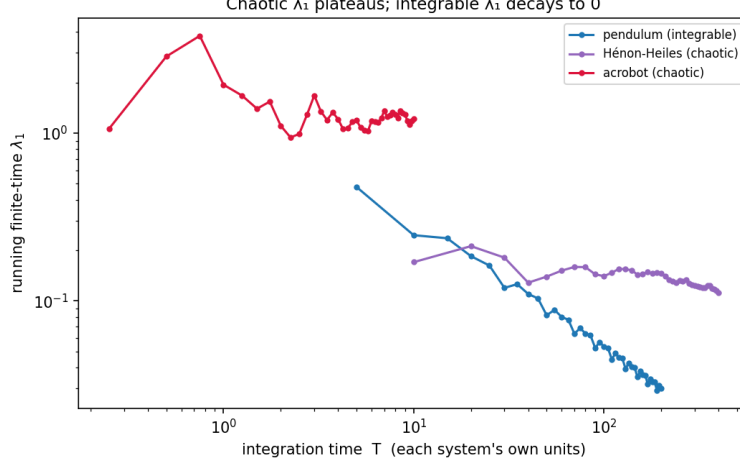


Figure 2: **Pre-correction.** Running finite-time λ_1 (Benettin) versus integration time. The chaotic acrobot and Hénon–Heiles settle onto positive plateaus; the integrable pendulum decays toward zero as $\ln T/T$. This plateau-versus-decay separates Hénon–Heiles’s small but genuine λ_1 from the pendulum’s finite-time floor, independently of magnitude.

at slightly different frequencies. A perturbation that shifts the energy therefore moves onto an orbit that drifts steadily out of phase, the lag growing *linearly* in time; that growing lag is what $\|M_T\|_2$ measures, so $\|M_T\|_2 \propto T$.¹ The rate is the *non-isochronicity*—the change of period with energy, $P'(E)$ —which vanishes for an isochronous system such as the harmonic oscillator, whose gain stays bounded. Because the growth is only linear, $\lambda_1 = \lim_{T \rightarrow \infty} T^{-1} \ln \|M_T\|_2 = 0$ exactly; a finite-horizon slope returns not zero but a small positive number *decaying* with the window as $\ln T/T$ ($0.05\text{--}0.08\text{ s}^{-1}$ over 40 s here)—a finite-time artifact of fitting an exponential to a line, not a Lyapunov exponent. This decay-versus-plateau is itself the cleanest chaos test: Figure 2 tracks the running λ_1 against integration time, and it settles onto a positive constant for the two chaotic systems (acrobot ≈ 1.2 , Hénon–Heiles ≈ 0.11) while sliding steadily toward zero for the pendulum. Hénon–Heiles’s small exponent is thus a genuine constant, not a slow artifact like the pendulum’s floor—a distinction of trend, not of magnitude.

Two points fix the chaotic case. First, the rate is a *single* factor of λ_1 : we fit $\|M_T\|_2$, which grows as $e^{\lambda_1 T}$, so its log-slope is λ_1 ; a squared quantity—a squared-loss objective, or the squared separation $\|\delta x_T\|^2$ —would grow as $e^{2\lambda_1 T}$ and read $2\lambda_1$. Second, the coincidence of slope and Benettin exponent is the *consistency relation* of Section 3.3, not an identity: the two are exact equals only for a constant-Jacobian map, and on a single acrobot orbit they agree only to within the finite-time scatter of the exponent (here $\sim 10\%$). The scatter-controlled statement $\text{slope} = \lambda_1$ across the whole integrable–chaotic range is established in Section 4.2.

4.2 The law holds across regimes

Section 4.1 illustrated the law with historical measurements; here we vary the dynamics using the corrected simulator. We sweep the acrobot across initial energies (θ_1 from 0.5 to 3.0, with

¹In action–angle coordinates (I, φ) the action I is conserved and the angle advances as $\varphi_T = \varphi_0 + \omega(I)T$, so the variational matrix is the pure shear $\begin{pmatrix} \omega'(I) & 1 \\ 0 & 1 \end{pmatrix}$: area-preserving ($\det = 1$), top singular value $\sigma_{\max} \sim |\omega'(I)|T$, with $\omega = 2\pi/P$.

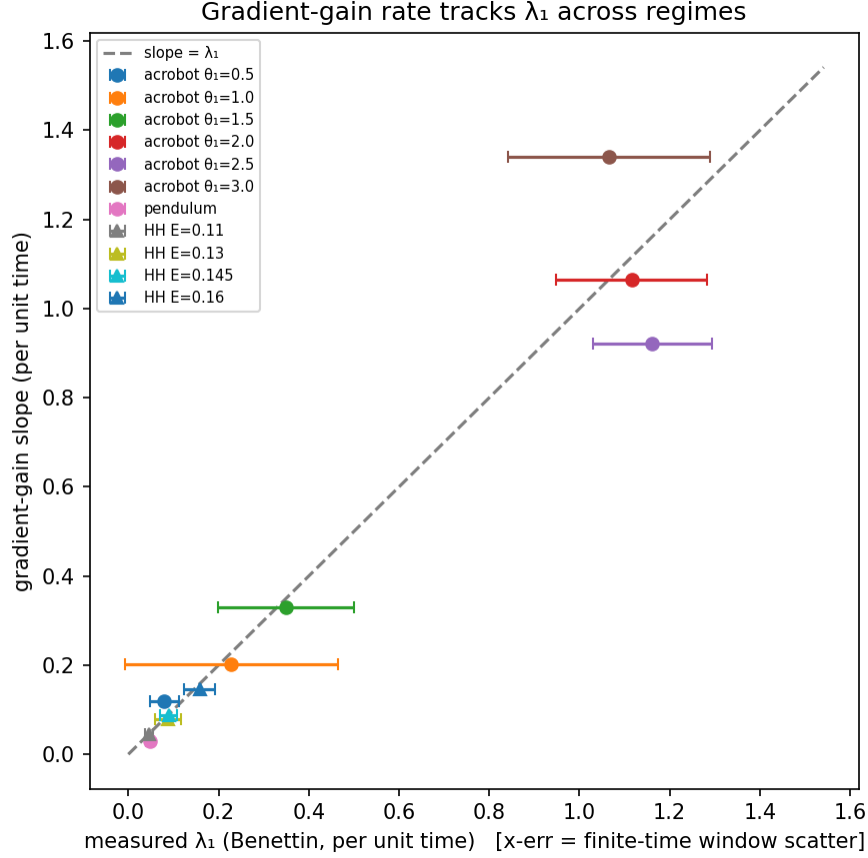


Figure 3: Gradient-gain slope versus Benettin λ_1 across an acrobot energy sweep ($\theta_1 = 0.5 \rightarrow 3.0$), the pendulum, and a Hénon–Heiles energy sweep (a second, independent chaotic system; triangles); all cluster on $y = x$ (correlation 0.969). Units are each system’s own inverse time (mechanical systems in s^{-1} , Hénon–Heiles dimensionless), so $y = x$ is the unit-free statement slope = λ_1 . Horizontal bars are finite-time Lyapunov window scatter.

λ_1 spanning from near 0.080 to $1.161 s^{-1}$, non-monotonically), add the integrable pendulum, and include a second, mechanistically independent chaotic system—Hénon–Heiles, swept across its own energies. For each we compare the gradient-gain slope against the Benettin λ_1 (Fig. 3). The points cluster on the line slope = λ_1 : across the integrable-to-chaotic range the two rates correlate at 0.969, with best-fit slope 0.986 (ideal 1) and intercept -0.001 . The gradient-gain rate thus approximately tracks the finite-time Lyapunov estimate across these regimes.

The 11-point sweep uses six acrobot states $(\theta_1, 0, 0, 0)$ with $\theta_1 \in \{0.5, 1, 1.5, 2, 2.5, 3\}$, pendulum state $(2, 0)$, and four Hénon–Heiles energies 0.11, 0.13, 0.145, 0.16. Measurement horizons are 1, 2, $\dots$, 8 seconds for the acrobot ($dt = 0.0005$ seconds), 10, 20, $\dots$, 80 seconds for the pendulum, and 20, 40, $\dots$, 120 time units for Hénon–Heiles.

The comparison also includes a second, mechanistically independent chaotic system: the Hénon–Heiles orbits (a smooth galactic potential, λ_1 up to ~ 0.15) fall on the same line slope = λ_1 , supporting the relation beyond the acrobot’s particular mechanics. The horizontal error bars report the finite-time scatter of the Benettin estimate over independent windows—an honest depiction of

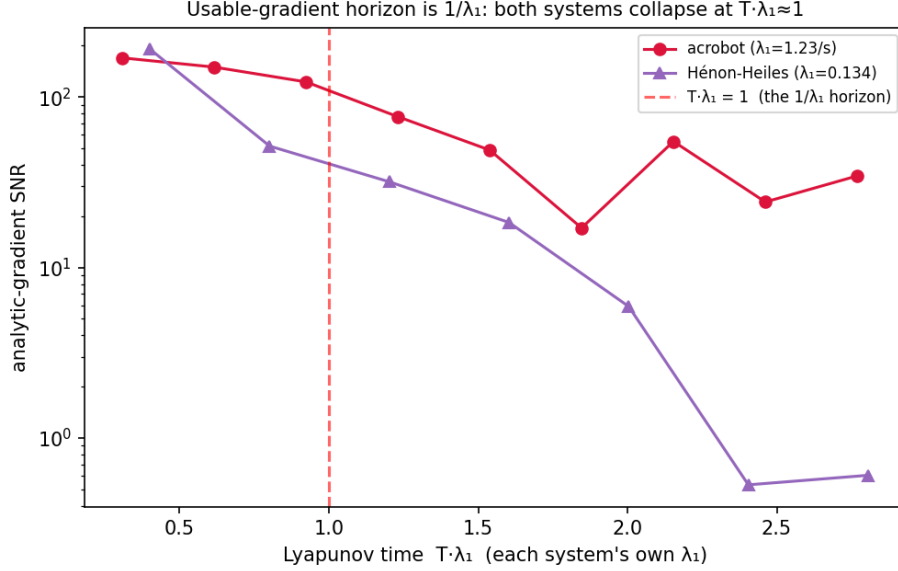


Figure 4: **Pre-correction.** Analytic-gradient SNR versus Lyapunov time $T\lambda_1$ for two chaotic systems whose λ_1 differ tenfold (acrobot 1.23 s^{-1} , Hénon–Heiles 0.13), each on its own λ_1 . Both collapse at the same $T\lambda_1 \approx 1$ (dashed): the gradient’s direction decorrelates across nearby initial conditions even as its magnitude keeps growing, so the usable-gradient horizon is $1/\lambda_1$ —set by the Lyapunov clock, not the rollout length.

finite-time Lyapunov uncertainty, not a confidence interval on the plotted point.

4.3 The gradient predictability horizon (pre-correction)

The growth law has a sharp operational consequence. Section 2.2 argued that because $\|M_T\|$ inflates both a gradient and its variance, beyond $T \sim 1/\lambda_1$ the analytic gradient of a differentiable rollout is dominated by chaotic amplification rather than by the objective—the *curse of chaos*. We make this quantitative with a signal-to-noise ratio: perturbing the initial state over an ensemble of nearby conditions, we compare the mean analytic gradient $\nabla_{x_0} \|x_T\|^2$ to its spread across the ensemble (Fig. 4). On the acrobot ($\lambda_1 = 1.23\text{ s}^{-1}$) the SNR is large well inside the horizon (≈ 170 at $T\lambda_1 \approx 0.3$) and falls by roughly an order of magnitude past $T\lambda_1 \approx 1$ (to ≈ 20 – 50): the gradient still has a large magnitude, but its *direction* is no longer shared across nearby initial conditions. The decisive point is that this crossover tracks the Lyapunov clock, not the rollout length. Hénon–Heiles, whose λ_1 is an order of magnitude smaller (0.13), collapses at the *same* place on its own Lyapunov axis—from ≈ 190 to below 1 through $T\lambda_1 \approx 1$. Two systems that differ tenfold in λ_1 thus lose gradient usability at the same $T\lambda_1$: the usable-gradient horizon is $T \sim 1/\lambda_1$, up to a logarithmic factor $\sim \log(1/\varepsilon)/(\lambda_1 dt)$ set by the perturbation scale ε .

This collapse is genuinely nonlinear, not a normalisation artifact. Each ensemble gradient has magnitude $\sim \|M_T\| \sim e^{\lambda_1 T}$, so the ensemble variance grows as $e^{2\lambda_1 T}$; on a purely linear (isochronous) map this common growth cancels between the mean (signal) and the spread (noise) and the SNR stays flat. The observed decline therefore measures real loss of *directional* agreement among neighbouring trajectories: past $T \sim 1/\lambda_1$ the analytic gradient points a different way for every nearby start, which is exactly what makes long differentiable rollouts unusable for first-order

optimisation. Decorrelation by itself is not unique to chaos—even an integrable but *anharmonic* system loses coherence eventually, as neighbouring orbits at slightly different frequencies dephase—but that runs on the non-isochronicity clock $\sim 1/(\omega'\varepsilon)$, which recedes without bound as the ensemble tightens. What is chaos-specific is that the collapse is pinned to the Lyapunov clock at $T\lambda_1 \sim 1$: a horizon a practitioner cannot escape by shrinking ε .

4.4 Precision and horizon in trajectory optimisation

We test how the time horizon affects gradient-based trajectory optimisation. For each horizon T , we optimise a sequence of control inputs to minimise the error at time T , using 250 Adam iterations. We compare three tasks: matching a prescribed final state on the chaotic acrobot, matching a prescribed final state on the integrable pendulum, and bringing both acrobot links upright without constraining their final velocities. Each state-matching target is generated by a known control sequence at the corresponding horizon, ensuring that a solution exists.

Figure 5 reports the terminal error after optimisation divided by the error obtained by applying zero control over the same interval. A value below one indicates improvement over passive motion; a value above one means optimisation leaves a larger error. Each task and horizon has its own baseline. All three tasks use the same physical horizons, expressed on the horizontal axis in units of the acrobot’s Lyapunov time.

Precise acrobot reaching improves on its baseline in geometric mean through $T\lambda_1 = 3.5$, but its error rises to approximately 5, 52, and 45 times baseline at $T\lambda_1 = 4, 4.5$, and 5. Outcomes vary widely across targets: at four Lyapunov times, the 10th–90th percentile range spans 0.18–420 times baseline. Pendulum reaching and acrobot swing-up remain below their respective baselines in geometric mean at every tested horizon.

The contrast shows that long-horizon optimisation performance depends on the objective: precise acrobot state matching deteriorates while the less restrictive upright objective remains improvable. It does not establish a universal Lyapunov-time limit on control. Longer horizons also introduce more control variables, and this experiment does not separate sensitivity growth from the effects of optimiser settings and the fixed iteration budget.

All tasks use matched time steps and horizons, with $dt = 0.0005$ s. Results summarise 12 acrobot reach targets, six pendulum targets, and six swing-up initialisations using geometric means and 10th–90th percentile bands.² The horizontal axis uses the acrobot reference $\lambda_1 = 1.00656 \text{ s}^{-1}$, measured by QR over seven seconds after discarding the first second of an eight-second passive rollout ($k = 4$, QR seed 0).

5 Part B — Lyapunov fidelity of learned world models

Part A established that the largest Lyapunov exponent λ_1 governs the gradients a differentiable simulator produces when the physics is correct. Part B asks the dual question for a model that is *learned* rather than given: does a world model trained only to predict the next state inherit the right λ_1 ? This is the spine of the section. A multilayer perceptron (MLP) fit by next-step regression

²Canonical acrobot start (2.5, 0, 0, 0); pendulum start (2, 0). Tested horizons are $T\lambda_1 = 1, 2, 3, 3.5, 4, 4.5, 5$. Adam learning rates are 0.3 for reaching and 1.0 for swing-up. Reach controls start at zero; swing-up controls start from seeded Gaussian noise with standard deviation 0.3. Acrobot control applies a generalised force to p_2 . Swing-up adds $10^{-7} \sum_t u_t^2$ to the terminal angle cost during optimisation; the plotted cost excludes this regulariser. No balancing or small terminal velocity is required.

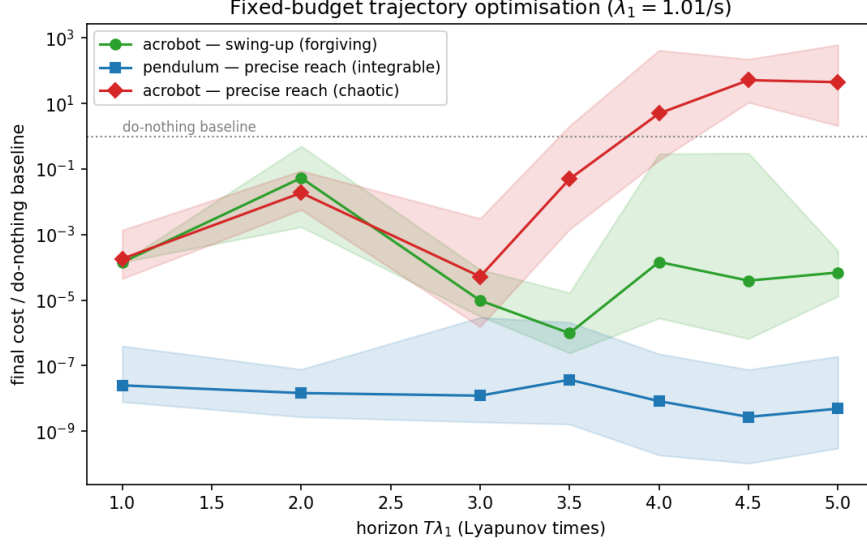


Figure 5: Fixed-budget trajectory optimisation: precise acrobot reaching (red), pendulum reaching (blue), and acrobot swing-up (green). Cost is normalised by each objective’s zero-control baseline (dotted line); lower is better. Curves show geometric means and shaded bands the 10th–90th percentiles across targets or initialisations. All curves share the acrobot’s Lyapunov-time axis, $\lambda_1 = 1.00656 \text{ s}^{-1}$.

reproduces the integrable exponent but *over-amplifies* the chaotic one in the historical comparison below. The corrected five-seed study compares soft and hard structural constraints: the HNN gives more consistent estimates across seeds, while the selected penalty model has a slightly closer median to the reference.

We reuse the cast of Part A: the integrable *pendulum* as the null contrast and the chaotic *acrobot* as the workhorse. The argument runs in four steps: the learned model and how fast its rollout drifts from the truth (Section 5.1); the learned-versus-true λ_1 that its Jacobian implies (Section 5.2); why next-step error alone does not certify λ_1 (Section 5.3); and how structural constraints affect it (Section 5.4).

5.1 The learned world model and its prediction drift (pre-correction)

A world model replaces the simulator’s one-step map $x_{t+1} = F(x_t)$ with a learned surrogate g_ϕ , fit so that iterating it reproduces trajectories of the true system. We deliberately use the smallest model that exhibits the effect: a residual multilayer perceptron $g_\phi(x) = x + \text{net}_\phi(x)$ with two tanh hidden layers, trained by next-step mean-squared error on rollouts of the true simulator. It is given only the state pairs (x_t, x_{t+1}) —no equations of motion, no Hamiltonian, no conserved quantity—which is exactly the regime in which one hopes a generic learner will simply absorb the dynamics from data.

Such a model can be accurate one step at a time yet useless over a rollout, so the first question is how quickly its predictions drift. We roll g_ϕ forward from a shared initial condition and track the prediction error $\|\hat{x}_t - x_t\|$ against the true trajectory (Fig. 6). On both systems the error grows, but at visibly different rates: on the integrable pendulum it rises slowly, while on the chaotic acrobot it climbs roughly twice as fast before saturating. A log-linear fit over the pre-saturation window gives

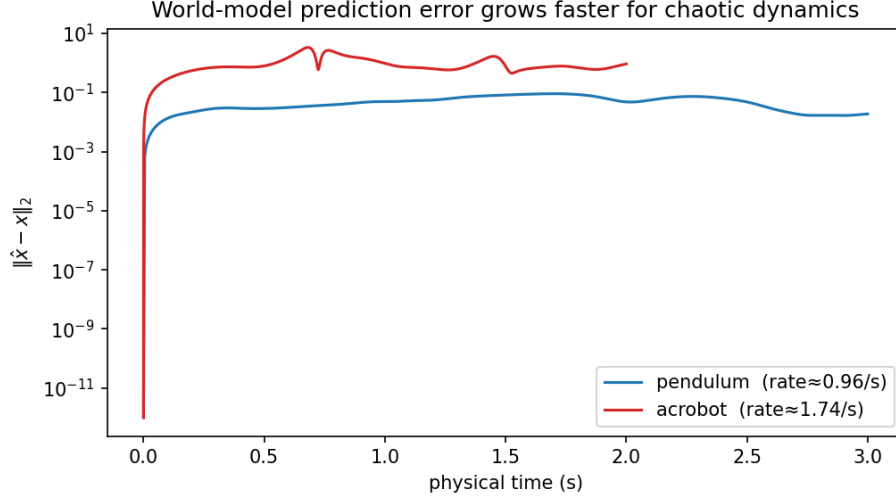


Figure 6: **Pre-correction.** Residual-MLP world-model prediction error $\|\hat{x}_t - x_t\|$ versus physical time (semilog). Error grows on both systems but faster on the chaotic acrobot (effective log-slope 1.74s^{-1}) than on the integrable pendulum (0.96s^{-1}). These are model-fidelity rates, not λ_1 : the pendulum’s is far from its true $\lambda_1 \approx 0$. The Lyapunov comparison is Section 5.2.

effective rates of 0.96s^{-1} (pendulum) and 1.74s^{-1} (acrobot).

These rates measure *model fidelity*, not the Lyapunov exponent. Two effects are superimposed: the model’s own approximation error, which accumulates even on the non-chaotic pendulum, and—on the acrobot—the genuine chaotic amplification of that error. The pendulum makes the distinction concrete: its fitted 0.96s^{-1} is nowhere near its true $\lambda_1 \approx 0$, so the drift rate cannot be read as λ_1 . Figure 6 is therefore a qualitative statement—prediction degrades faster on chaotic dynamics—and isolating the exponent the model has actually learned requires probing its Jacobian directly, the subject of Section 5.2.

5.2 Learned versus true λ_1 (pre-correction)

To read off the exponent the model has learned, we apply the *same* Benettin/QR estimator used for the true systems (Section 3.2) to the model’s one-step Jacobian $\partial g_\phi / \partial x$, obtained by automatic differentiation. The only subtlety is where to evaluate it: rolling the model out and estimating λ_1 along its *own* trajectory would conflate an inaccurate Jacobian with a drifted orbit, so we instead evaluate the learned Jacobian along the *true* trajectory, after the same transient. Estimator, orbit, and transient are then identical for the learned and reference exponents; only the source of the Jacobian differs—true physics versus learned model—so any gap is the model’s Jacobian alone. A faithful model, whose Jacobians match the true ones along the orbit, would return the true λ_1 .

Figure 7 plots learned against true λ_1 for the two systems. On the integrable pendulum the model lands on the truth: a learned $\lambda_1 \approx 0.046\text{s}^{-1}$ against a true 0.063s^{-1} , both effectively zero—the network correctly reports non-chaotic dynamics. On the chaotic acrobot it does not: the learned exponent is 3.45s^{-1} against a true 1.04s^{-1} , an over-amplification by a factor of 3.3. The model is not merely imprecise here; it manufactures sensitivity the true dynamics do not have.

The asymmetry is the point. The network is trained to low next-step error on both systems, yet it

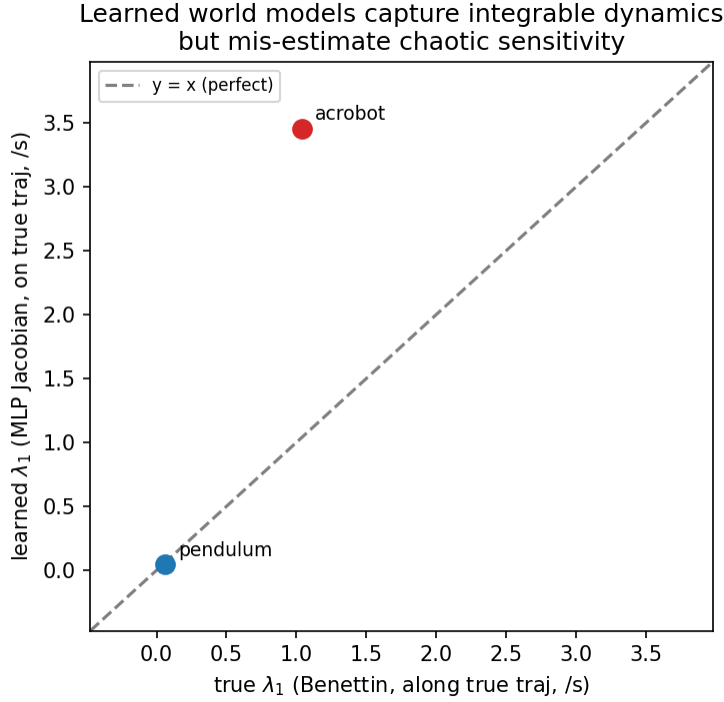


Figure 7: **Pre-correction.** Learned versus true λ_1 , both from the same Benettin/QR estimator evaluated along the true orbit. The integrable pendulum sits on the $y = x$ line (both $\lambda_1 \approx 0$); the chaotic acrobot lies far above it—learned 3.45 s^{-1} versus true 1.04 s^{-1} , an over-amplification of $3.3\times$. Next-step accuracy does not imply the right λ_1 .

reproduces λ_1 only where the exponent is trivial and inflates it precisely where chaotic sensitivity makes it matter. Low next-step error is therefore no guarantee of dynamical fidelity: λ_1 is a property the training objective never sees, and on chaotic dynamics the learned model overstates it threefold. Section 5.3 explains why next-step regression is blind to it.

5.3 Why next-step error does not certify sensitivity

The comparison in Section 5.2 motivates a distinction between prediction and sensitivity. Next-step mean-squared error penalises $\|g_\phi(x_t) - x_{t+1}\|^2$ at the training states: it supervises *where* points are mapped. But λ_1 is a property of the *Jacobian* $\partial g_\phi / \partial x$, which governs how infinitesimal perturbations are stretched, and next-step error does not directly target a derivative. With finite data and finite capacity a model can drive its pointwise error low while its Jacobian—constrained by no derivative target—remains substantially wrong. The threefold over-amplification of Section 5.2 is exactly that gap made visible: the network reproduces the trajectory well yet equips it with the wrong sensitivity.

There is a sharper way to name what is missing. The true dynamics are Hamiltonian, and in canonical coordinates they conserve phase-space volume exactly—the Lyapunov spectrum sums to zero (Eq. (6)). That volume law is a structural fingerprint of genuine mechanics, yet it is a property of the Jacobian’s *determinant*—an object next-step regression does not directly supervise—so low pointwise error alone does not certify it. The corrected study compares λ_1 in the same canonical coordinates for all models and the reference, avoiding finite-window coordinate changes. The

same reasoning motivates building structure in. Section 5.4 contrasts a soft penalty for volume preservation with a Hamiltonian architecture and symplectic integration. Volume preservation constrains the spectrum sum, but does not determine the largest exponent.

5.4 Structure-preserving world models

Does building physical structure into a world model improve its estimate of the simulator’s sensitivity? We compare three models of the chaotic acrobat: a plain MLP trained to predict the next state; an MLP with an additional penalty encouraging phase-space volume preservation; and a Hamiltonian neural network (HNN), which learns a Hamiltonian and advances the state using implicit midpoint integration. Volume preservation constrains the sum of the Lyapunov exponents, but does not by itself determine the largest exponent. The volume penalty adds $\mu(\log|\det \partial g_\phi/\partial x|)^2$ to the next-step loss, encouraging a unit Jacobian determinant.

The HNN represents the Hamiltonian explicitly. It learns a scalar Hamiltonian $H_\phi(q, p)$ on canonical coordinates ($q = \theta$, $p = M(\theta)\omega$, with M the mass matrix), forms the Hamiltonian flow $\dot{q} = \partial H/\partial p$, $\dot{p} = -\partial H/\partial q$, and advances it with an implicit midpoint step; angles enter through a $(\cos \theta, \sin \theta)$ embedding to respect the circle topology. This step is symplectic when solved exactly; the finite-iteration map is checked against its analytic midpoint Jacobian. Training uses midpoint Hamiltonian-gradient targets.

We train each model with five random seeds while keeping the training dataset fixed. For evaluation, we apply each model’s Jacobian along the same reference simulator trajectory and estimate the resulting perturbation-growth rate. This compares local sensitivity along a shared trajectory; it does not measure the divergence of trajectories generated independently by each model.

The HNN gives substantially more consistent estimates across training seeds: its estimates range from 0.9504 to 1.0280 s^{-1} , compared with 1.0035 – 1.6348 for the plain MLP and 0.8889 – 1.5178 for the selected penalty MLP. However, all five HNN estimates fall below the reference of 1.11634 s^{-1} . The selected penalty model has a slightly closer median, although its penalty weight was chosen using this same evaluation. Figure 8 reports medians of 1.2790 , 1.2280 , and 0.9871 s^{-1} for the plain MLP, penalty MLP, and HNN, respectively. The penalty weight $\mu = 10$ was selected from $\{0.1, 1, 10\}$ by median proximity to the reference, not on an independent validation set.

The three model conditions shown in Figure 8 each use training seeds 0–4, all with dataset seed 0. Results are saved in `writeup/figures/partb_ablation.json`. Each dataset contains 80 trajectories of 2000 steps. The MLPs train for 120 epochs and the HNNs for 200, with Adam (learning rate 10^{-3} , batch size 256). This compares the existing model protocols, not equal training-compute budgets. Each exponent averages eight QR initialisations with $k = 4$ over 54 seconds, $dt = 0.0005$ seconds, after discarding 1200 trajectory steps (0.6 seconds) from 109200 steps beginning at canonical state $(2.5, 0, 0, 0)$. The reference’s full QR-initialisation range is 0.07296 s^{-1} .

The main finding is therefore greater consistency across training seeds for the Hamiltonian model, rather than uniformly greater accuracy. Physical structure reduces the observed variation in this comparison, but does not guarantee the correct sensitivity.

6 Extensions

The diagnostics here are deliberately demonstrated on low-dimensional, single-threaded CPU systems. The natural next steps push them onto GPU hardware and onto the large learned models of the

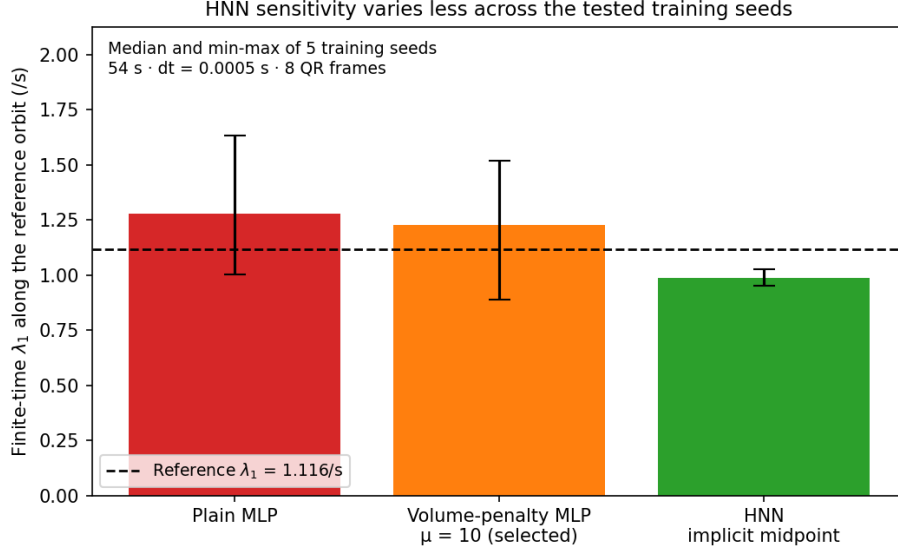


Figure 8: Saved five-seed study: median and min-max of plain MLP, selected penalty MLP ($\mu = 10$), and corrected midpoint HNN, with matching reference 1.11634 s^{-1} . Protocol: 54 seconds, $dt = 0.0005$ seconds, eight QR frames, $k = 4$, canonical $(2.5, 0, 0, 0)$; learned Jacobians follow the reference trajectory. Whiskers reflect training seeds, not confidence intervals.

surrounding Physical-AI stack.

- **Scale to GPU.** Every diagnostic in this paper—the Benettin/QR estimator, the gradient-gain slope, and the perturbed-rollout SNR—is a sweep or ensemble average over many *independent* trajectories (initial-condition averaging, the energy sweep, the SNR ensemble). The outer loop over trajectories, seeds, and sweep points is therefore *embarrassingly parallel* and maps directly onto the GPU—in the spirit of the thousands of parallel environments used for large-scale robot learning—with only the per-trajectory time-stepping remaining sequential. Re-implementing the Warp simulators on **Newton** (GPU; the production successor to `warp.sim`) is one possible path for these larger-scale extensions.
- **Cosmos / large world models.** The Part-B fidelity test extends from our small surrogate to a large video world model such as **Cosmos**: does it preserve the physical Lyapunov spectrum, or hallucinate (and wash out) sensitivity? The error-growth diagnostic (Section 5.1) transfers to pixel space as a qualitative probe; the full Lyapunov-fidelity diagnostic (Section 5.2) requires recovering physical state (θ, ω) from generated video—a perception front-end—or a latent-space tangent-vector adaptation, both research problems rather than a front-end swap, and both demanding GPU-scale compute.
- **GR00T / action models.** For a closed-loop policy such as **GR00T**, the natural robustness diagnostic is the finite-time Lyapunov exponent of the observation $\rightarrow$ action $\rightarrow$ next-state map, with the policy in a simulation loop (Isaac Lab / Newton) and numerical (perturbed-rollout) sensitivities, since a large policy with a finite context window is a non-Markovian, non-autodiff-friendly map.
- **Contact and stiff dynamics.** A distinct gradient-pathology regime—non-smooth rather than chaotic—deliberately outside this conservative CPU study, but the setting where differentiable-

simulator gradients most often break in practice.

7 Limitations

All reported exponents are finite-window estimates. Window length, QR initialisation, coordinates, discretisation and initial state matter; this study does not establish asymptotic convergence for every condition. The five-seed model comparison holds the dataset fixed and does not measure data-sampling variation. Penalty selection reuses the evaluation, and reference-itinerary Jacobians do not establish autonomous-rollout fidelity. Volume preservation alone cannot determine the largest exponent. The optimisation comparison uses one fixed optimiser budget and does not establish a universal impossibility boundary. The explicitly historical figures use an obsolete acrobot implementation. These low-dimensional CPU results do not validate Cosmos, video world models, GPU performance, or real-robot behaviour.

A The Benettin/QR estimator: telescoping and the log-sum

Sections 2.1 and 3.2 assert that averaging the logarithms of the QR diagonals recovers the spectrum; we record here why, since the running sum is never formed explicitly in the main text.

Let $J_0, \dots, J_{N-1}$ be the one-step Jacobians along an orbit and $Q^{(0)} \in \mathbb{R}^{n \times k}$ the initial orthonormal frame ($Q^{(0)\top} Q^{(0)} = I_k$). Each step performs a thin QR factorisation and feeds the resulting frame to the next step,

$$J_t Q^{(t)} = Q^{(t+1)} R^{(t+1)}, \quad t = 0, \dots, N-1, \quad (11)$$

with $Q^{(t+1)}$ orthonormal-columned and $R^{(t+1)}$ upper triangular with positive diagonal. Substituting (11) repeatedly telescopes the rollout Jacobian $M_N = \prod_{t < N} J_t$ acting on the initial frame,

$$M_N Q^{(0)} = J_{N-1} \cdots J_0 Q^{(0)} = Q^{(N)} \underbrace{R^{(N)} R^{(N-1)} \cdots R^{(1)}}_{\mathcal{R}_N}. \quad (12)$$

Equation (12) is itself a thin QR factorisation of $M_N Q^{(0)}$: the intermediate frames are not cancelled but carried forward as each step's input, and the entire accumulated rotation collects in the single factor $Q^{(N)}$. Two properties then isolate the spectrum.

First, $\mathcal{R}_N$ is a product of upper-triangular matrices, hence upper triangular with diagonal equal to the product of diagonals,

$$(\mathcal{R}_N)_{ii} = \prod_{t=1}^N R_{ii}^{(t)} \implies \ln(\mathcal{R}_N)_{ii} = \sum_{t=1}^N \ln R_{ii}^{(t)}, \quad (13)$$

converting the overflow-prone matrix product into the additive running sum the estimator accumulates, the numerical reason for re-orthonormalising at every step rather than forming M_N once.

Second, $Q^{(N)}$ has orthonormal columns and is therefore an isometry: $\|Q^{(N)}x\| = \|x\|$, its singular values are all one, $\|Q^{(N)}\| = 1$, and so $\sigma_i(M_N Q^{(0)}) = \sigma_i(\mathcal{R}_N)$. It rotates the frame but neither stretches nor shrinks it, contributing nothing to any growth rate. Dividing (13) by the elapsed physical time $N dt$ and letting $N \rightarrow \infty$ gives

$$\lambda_i = \lim_{N \rightarrow \infty} \frac{1}{N dt} \sum_{t=1}^N \ln R_{ii}^{(t)}, \quad (14)$$

with convergence guaranteed by Oseledets’ theorem for a generic $Q^{(0)}$. “Generic” is what the seeded *random* frame of Section 3.2 secures: with probability one the columns of $Q^{(0)}$ avoid the measure-zero subspaces of the Oseledets filtration that grow more slowly than the top- k rates, so (14) returns $\lambda_1 \geq \dots \geq \lambda_k$ rather than a slower set. That the *diagonal* of $\mathcal{R}_N$ suffices—rather than its singular values—is exactly because the isometry $Q^{(N)}$ ties the two together.

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